

CS 171 / EE 142: Intro to ML and DM

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Slide Set 17: Neural Network Learning



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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 - ▶ For use only by enrolled students in the course

Neural Network Learning

To learn a neural network, we use a standard loss function:

$$L = \sum_{i=1}^m l(y_i, f(\vec{x}_i))$$

$$l(y, f) = (y - f)^2 \quad \text{least squares regression}$$

$$l(y, f) = -\ln \sigma(yf) \quad \text{binary classification}$$

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But, the parameters are not a vector! They are a set of matrices!

Never fear! We will just extend the idea of a gradient:

$$\nabla_{\mathbf{W}} L = \begin{bmatrix} \frac{\partial L}{\partial w_{1,1}} & \frac{\partial L}{\partial w_{1,2}} & \cdots & \frac{\partial L}{\partial w_{1,d_2}} \\ \frac{\partial L}{\partial w_{2,1}} & \frac{\partial L}{\partial w_{2,2}} & \cdots & \frac{\partial L}{\partial w_{2,d_2}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial w_{d_1,1}} & \frac{\partial L}{\partial w_{d_1,2}} & \cdots & \frac{\partial L}{\partial w_{d_1,d_2}} \end{bmatrix}$$

$$\mathbf{W} \leftarrow \mathbf{W} - \eta \nabla_{\mathbf{W}} L$$

Neural Network Gradient Descent

If we have P layers,

```
{ $\mathbf{W}^{(1)}, \mathbf{W}^{(2)}, \dots, \mathbf{W}^{(P)}$ }  $\leftarrow$  random values (near zero)
while not converged do
  for all  $k \in \{1, 2, \dots, P\}$  do
     $\mathbf{G}^{(k)} \leftarrow \nabla_{\mathbf{W}^{(k)}} L$ 
  for all  $k \in \{1, 2, \dots, P\}$  do
     $\mathbf{W}^{(k)} \leftarrow \mathbf{W}^{(k)} - \eta \mathbf{G}^{(k)}$ 
return { $\mathbf{W}^{(1)}, \mathbf{W}^{(2)}, \dots, \mathbf{W}^{(P)}$ }
```

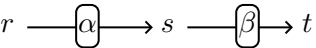
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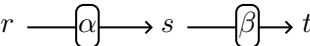
So, how to calculate $\nabla_{\mathbf{W}^{(k)}} L$?

Chain Rule for Scalars

$$t = \beta(\alpha(r))$$


```
graph LR; r --> alpha; alpha --> s; s --> beta; beta --> t
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$$\alpha(r) = s \quad \beta(s) = t$$

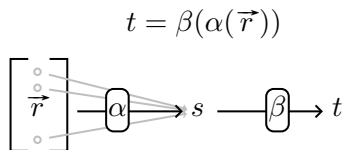
Chain Rule for Scalars

$$t = \beta(\alpha(r))$$
$$r \longrightarrow \boxed{\alpha} \longrightarrow s \longrightarrow \boxed{\beta} \longrightarrow t$$

$$\alpha(r) = s \quad \beta(s) = t$$

$$\frac{\partial s}{\partial r} \cdot \frac{\partial t}{\partial s} = \frac{\partial t}{\partial r}$$
$$\alpha'(r) \cdot \beta'(s) =$$

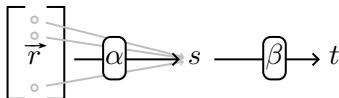
Chain Rule for Gradients



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Chain Rule for Gradients

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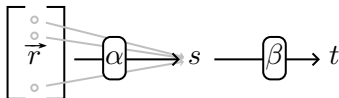


$$\alpha(\vec{r}) = s \quad \beta(s) = t$$

$$\begin{bmatrix} \frac{\partial t}{\partial r_1} \\ \frac{\partial t}{\partial r_2} \\ \vdots \\ \frac{\partial t}{\partial r_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial s}{\partial r_1} & \frac{\partial t}{\partial s} \\ \frac{\partial s}{\partial r_2} & \frac{\partial t}{\partial s} \\ \vdots & \vdots \\ \frac{\partial s}{\partial r_n} & \frac{\partial t}{\partial s} \end{bmatrix} = \nabla_{\vec{r}} t$$

Chain Rule for Gradients

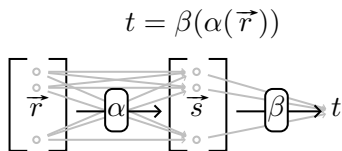
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$$\alpha(\vec{r}) = s \quad \beta(s) = t$$

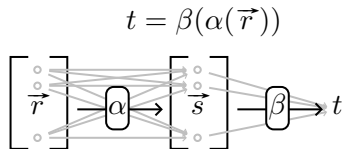
$$\begin{bmatrix} \frac{\partial t}{\partial r_1} \\ \frac{\partial t}{\partial r_2} \\ \vdots \\ \frac{\partial t}{\partial r_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial s}{\partial r_1} \beta'(s) \\ \frac{\partial s}{\partial r_2} \beta'(s) \\ \vdots \\ \frac{\partial s}{\partial r_n} \beta'(s) \end{bmatrix} = \nabla_{\vec{r}} t$$
$$\nabla \alpha(\vec{r}) \beta'(s) =$$

Chain Rule for Gradients



$$\alpha(\vec{r}) = \vec{s} \quad \beta(\vec{s}) = t$$

Chain Rule for Gradients



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$$\begin{bmatrix} \frac{\partial t}{\partial r_1} \\ \frac{\partial t}{\partial r_2} \\ \vdots \\ \frac{\partial t}{\partial r_n} \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^m \frac{\partial s_j}{\partial r_1} \frac{\partial t}{\partial s_j} \\ \sum_{j=1}^m \frac{\partial s_j}{\partial r_2} \frac{\partial t}{\partial s_j} \\ \vdots \\ \sum_{j=1}^m \frac{\partial s_j}{\partial r_n} \frac{\partial t}{\partial s_j} \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial s_1}{\partial r_1} & \frac{\partial s_2}{\partial r_1} & \cdots & \frac{\partial s_m}{\partial r_1} \\ \frac{\partial s_1}{\partial r_2} & \frac{\partial s_2}{\partial r_2} & \cdots & \frac{\partial s_m}{\partial r_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial s_1}{\partial r_n} & \frac{\partial s_2}{\partial r_n} & \cdots & \frac{\partial s_m}{\partial r_n} \end{bmatrix}}_{J_\alpha(\vec{r})^\top} \underbrace{\begin{bmatrix} \frac{\partial t}{\partial s_1} \\ \frac{\partial t}{\partial s_2} \\ \vdots \\ \frac{\partial t}{\partial s_m} \end{bmatrix}}_{\nabla \beta(\vec{s})}$$

$$J_{\alpha}(\vec{r}) = \begin{bmatrix} \frac{\partial \alpha_1(\vec{r})}{\partial r_1} & \frac{\partial \alpha_1(\vec{r})}{\partial r_2} & \cdots & \frac{\partial \alpha_1(\vec{r})}{\partial r_n} \\ \frac{\partial \alpha_2(\vec{r})}{\partial r_1} & \frac{\partial \alpha_2(\vec{r})}{\partial r_2} & \cdots & \frac{\partial \alpha_2(\vec{r})}{\partial r_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \alpha_m(\vec{r})}{\partial r_1} & \frac{\partial \alpha_m(\vec{r})}{\partial r_2} & \cdots & \frac{\partial \alpha_m(\vec{r})}{\partial r_n} \end{bmatrix}$$

Examples:

$$\alpha(\vec{r}) = \mathbf{M}\vec{r}$$

$\Downarrow$

$$J_{\alpha}(\vec{r}) = \mathbf{M}$$

$$\alpha(\vec{r}) = \tilde{g}(\vec{r})$$

$\Downarrow$

$$J_{\alpha}(\vec{r}) = \begin{bmatrix} g'(r_1) & 0 & \cdots & \cdots & 0 \\ 0 & g'(r_2) & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & g'(r_n) & 0 \end{bmatrix}$$

Hadamard (element-wise) Product

Note that

$$\begin{aligned} \begin{bmatrix} g'(r_1) & 0 & \dots & 0 \\ 0 & g'(r_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & g'(r_n) \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} &= \begin{bmatrix} g'(r_1)b_1 \\ g'(r_2)b_2 \\ \vdots \\ g'(r_n)b_n \end{bmatrix} \\ &= \begin{bmatrix} g'(r_1) \\ g'(r_2) \\ \vdots \\ g'(r_n) \end{bmatrix} \odot \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \\ &= \tilde{g}'(\vec{r}) \underbrace{\odot}_{\text{Hadamard product}} \vec{b} \end{aligned}$$

Neural Network Gradients

$$\begin{aligned}\nabla_{\mathbf{W}^{(k)}} L &= \nabla_{\mathbf{W}^{(k)}} \sum_{i=1}^m l(y_i, f(\vec{x}_i)) \\ &= \sum_{i=1}^m \nabla_{\mathbf{W}^{(k)}} l(y_i, f(\vec{x}_i))\end{aligned}$$

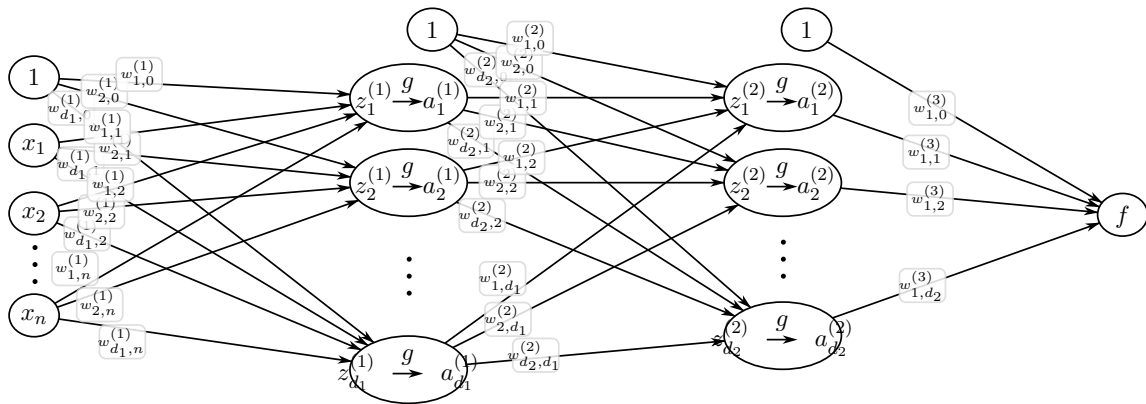
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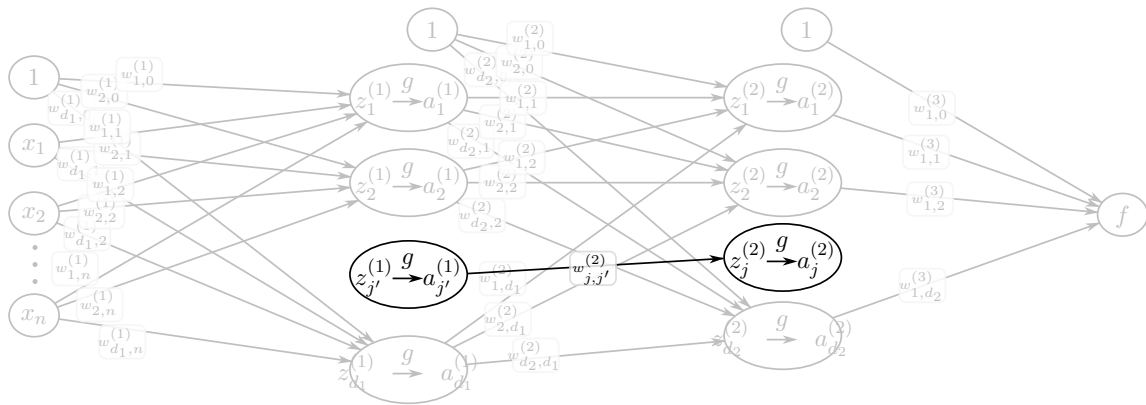
So, how to calculate $\nabla_{\mathbf{W}^{(k)}} l(y_i, f(\vec{x}_i))$?

(the derivatives of the loss for this example with respect to each weight in the network)

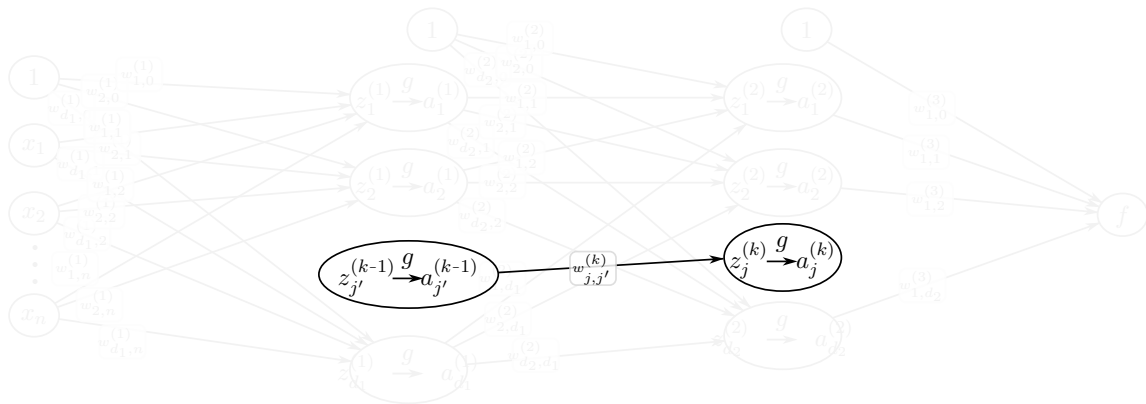
Weight gradients



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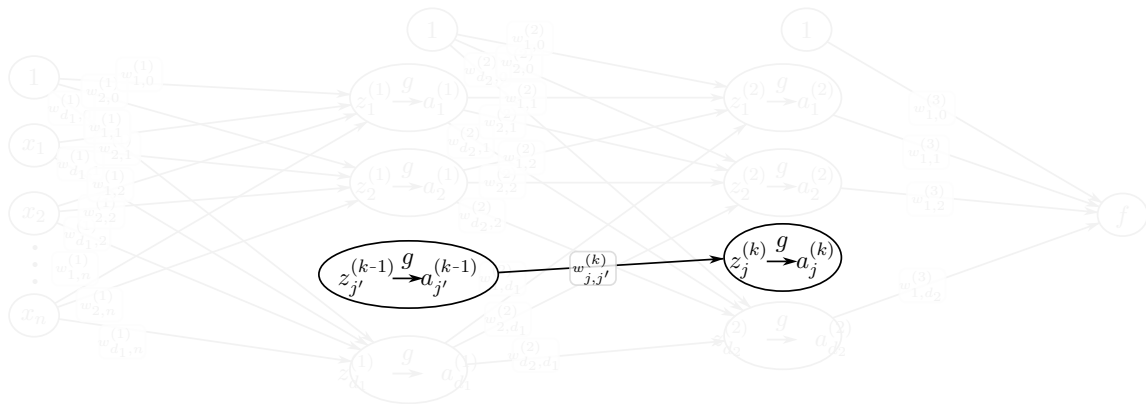


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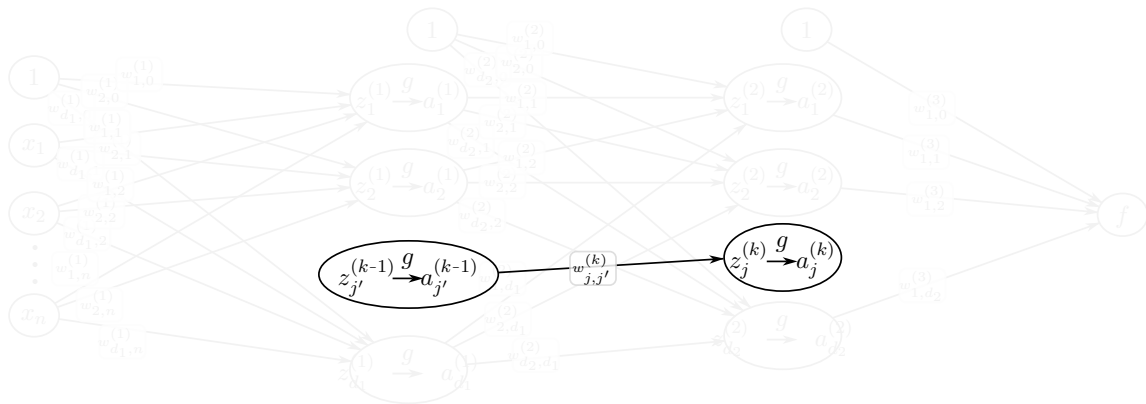
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$$\frac{\partial l}{\partial z_j^{(k)}} \frac{\partial z_j^{(k)}}{\partial w_{j,j'}^{(k)}} = \frac{\partial l}{\partial w_{j,j'}^{(k)}}$$



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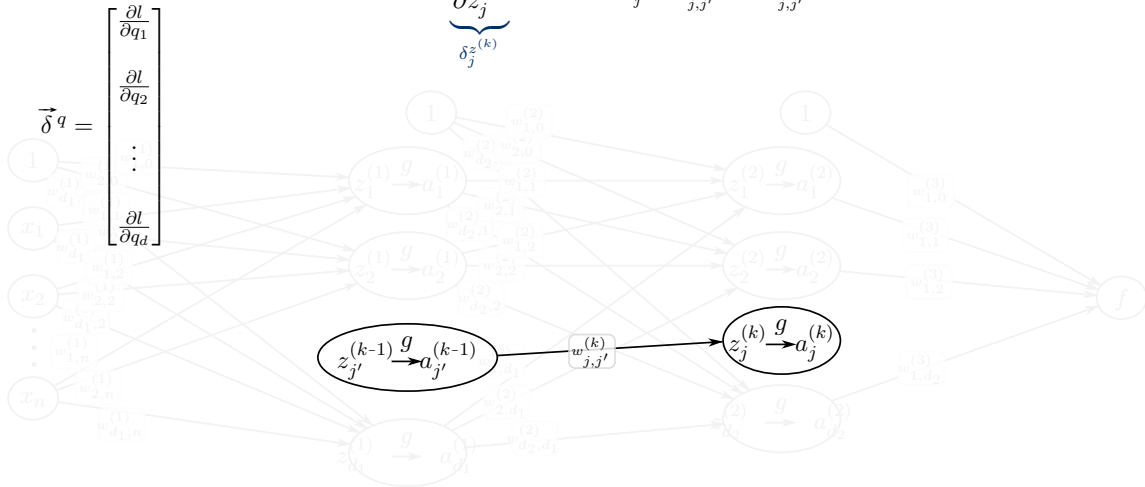
$$\frac{\partial l}{\partial z_j^{(k)}} a_{j'}^{(k-1)} = \frac{\partial l}{\partial z_j^{(k)}} \frac{\partial z_j^{(k)}}{\partial w_{j,j'}^{(k)}} = \frac{\partial l}{\partial w_{j,j'}^{(k)}}$$



Weight gradients

$$\vec{\delta}^q \triangleq \nabla_{\vec{q}} l$$

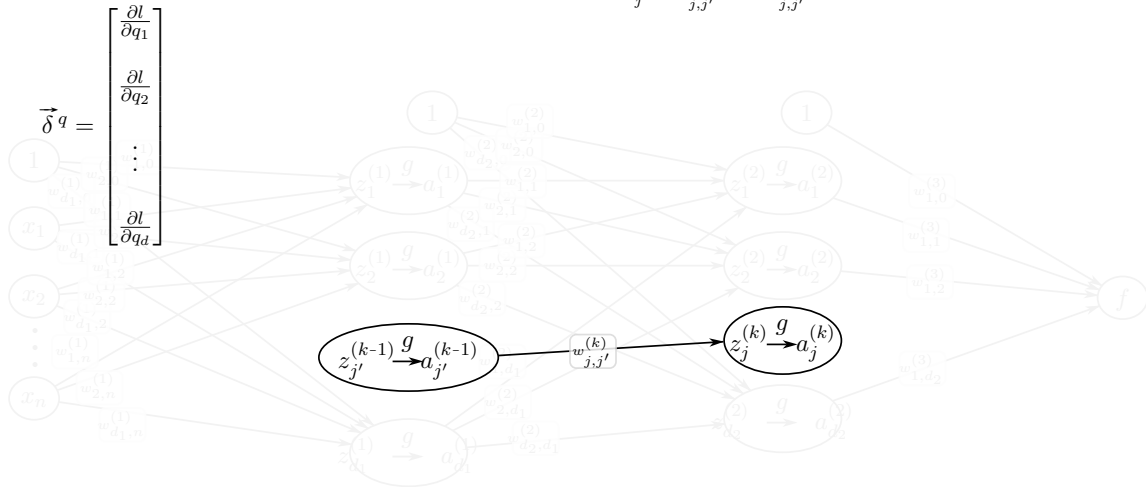
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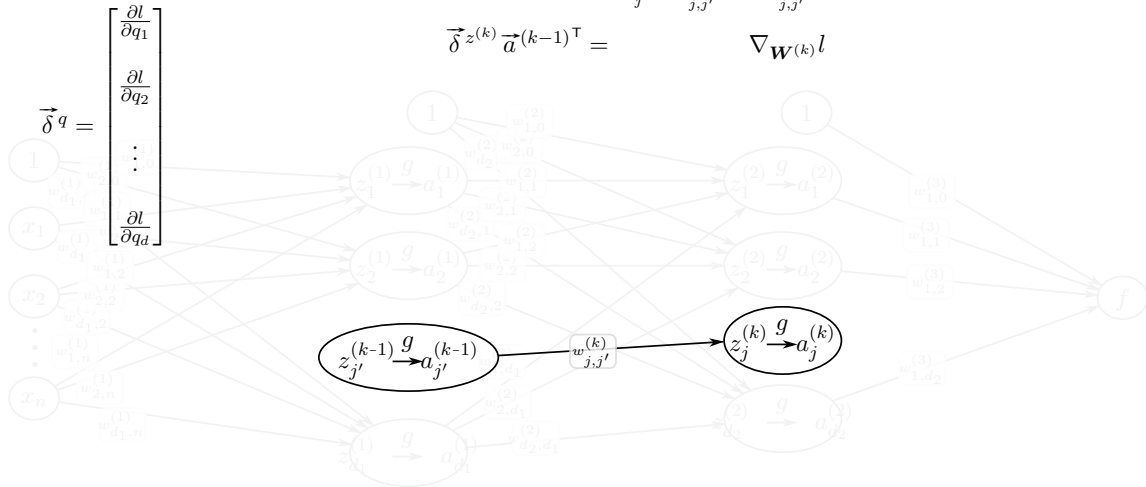


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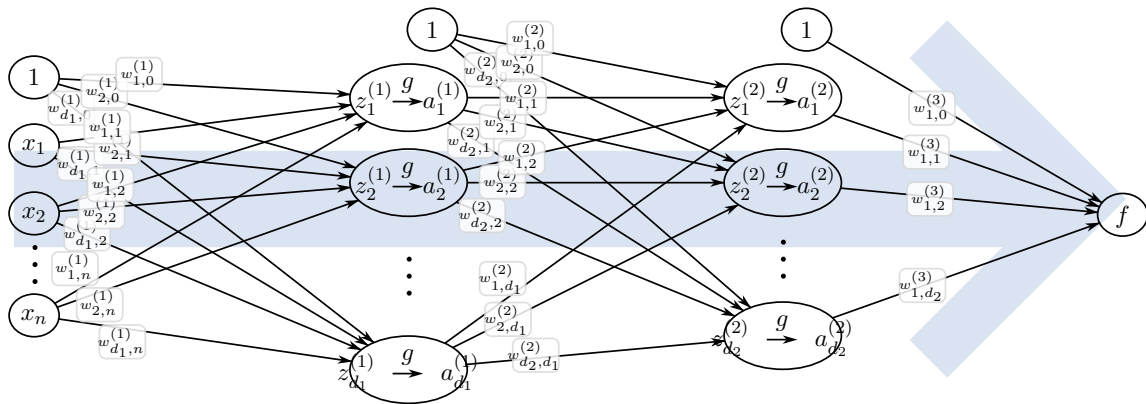
$$\vec{\delta}^{z^{(k)}} \vec{a}^{(k-1)\top} = \nabla_{\mathbf{W}^{(k)}} l$$



Simplified Diagram

forward propagation (calculates $f(\vec{x})$ from $\vec{x}$)

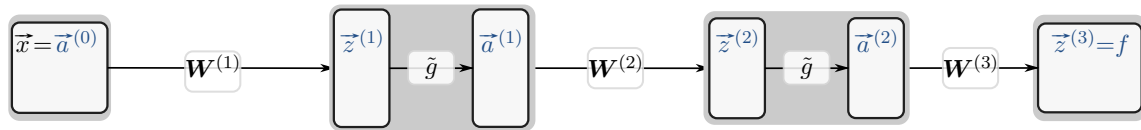
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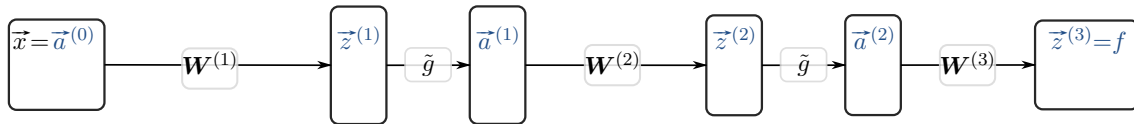
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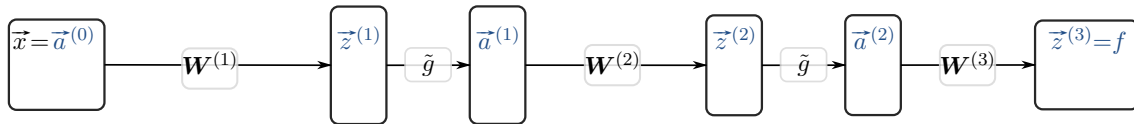
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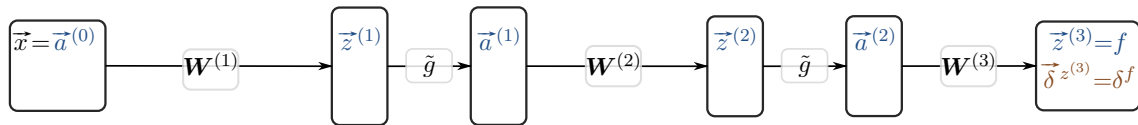
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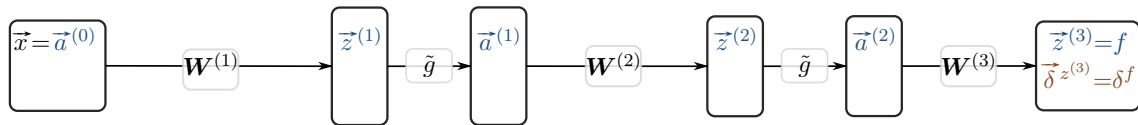
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$$\vec{\delta}^{z^{(3)}} = \left[\frac{\partial l(y_i, f)}{\partial f} \right]$$

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$$\vec{\delta}^{z^{(3)}} = \left[\frac{\partial l(y_i, f)}{\partial f} \right]$$

$$l(y, f) = (f - y)^2$$

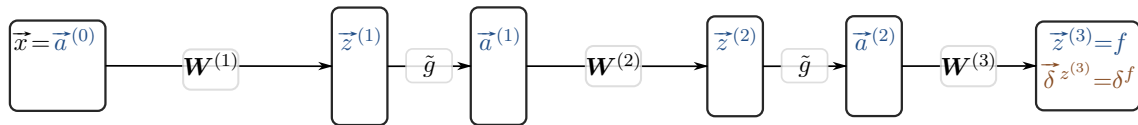
$$\frac{\partial l(y, f)}{\partial f} = 2(f - y)$$

$$l(y, \tilde{f}) = -\ln \sigma(y\tilde{f})$$

$$\frac{\partial l(y, \tilde{f})}{\partial \tilde{f}} = -(1 - \sigma(y\tilde{f}))y$$

Backward Propagation

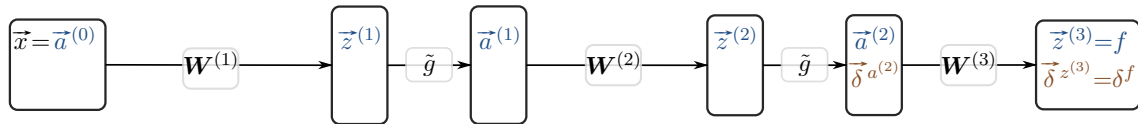
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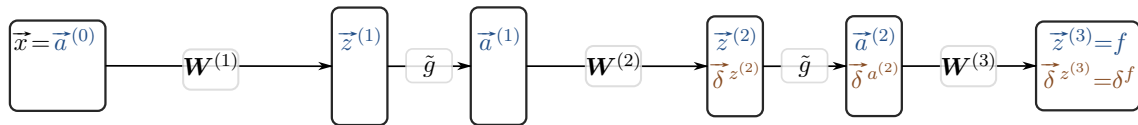


$$\vec{\delta} a^{(2)} = \mathbf{W}^{(3)\top} \vec{\delta} z^{(3)}$$

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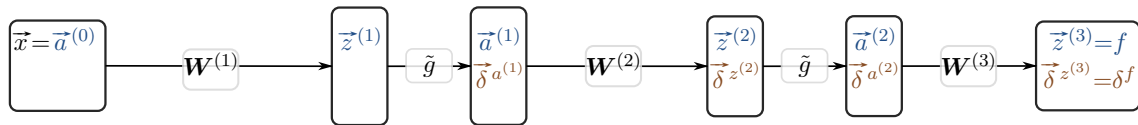


$$\vec{\delta} a^{(2)} = \mathbf{W}^{(3)\top} \vec{\delta} z^{(3)}$$

$$\vec{\delta} z^{(2)} = \tilde{g}'(\vec{z}^{(2)}) \odot \vec{\delta} a^{(2)} \quad \vec{\delta} z^{(3)} = \left[\frac{\partial l(y_i, f)}{\partial f} \right]$$

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$$\vec{x} = \vec{a}^{(0)} \quad \mathbf{W}^{(1)} \vec{a}^{(0)} = \vec{z}^{(1)} \quad \tilde{g}(\vec{z}^{(1)}) = \vec{a}^{(1)} \quad \mathbf{W}^{(2)} \vec{a}^{(1)} = \vec{z}^{(2)} \quad \tilde{g}(\vec{z}^{(2)}) = \vec{a}^{(2)} \quad \mathbf{W}^{(3)} \vec{a}^{(2)} = \vec{z}^{(3)} \quad \vec{z}^{(3)} = f$$



$$\vec{\delta} a^{(1)} = \mathbf{W}^{(2)\top} \vec{\delta} z^{(2)}$$

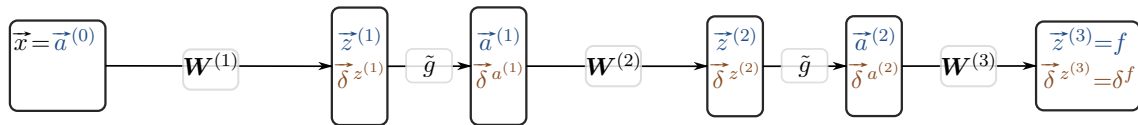
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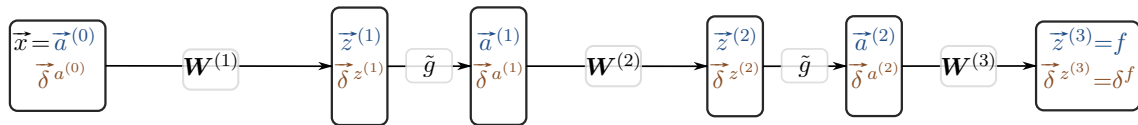
$$\vec{x} = \vec{a}^{(0)} \quad \mathbf{W}^{(1)} \vec{a}^{(0)} = \vec{z}^{(1)} \quad \tilde{g}(\vec{z}^{(1)}) = \vec{a}^{(1)} \quad \mathbf{W}^{(2)} \vec{a}^{(1)} = \vec{z}^{(2)} \quad \tilde{g}(\vec{z}^{(2)}) = \vec{a}^{(2)} \quad \mathbf{W}^{(3)} \vec{a}^{(2)} = \vec{z}^{(3)} \quad \vec{z}^{(3)} = f$$



$$\begin{aligned} \vec{\delta}^{a(1)} &= \mathbf{W}^{(2)\top} \vec{\delta}^{z(2)} & \vec{\delta}^{a(2)} &= \mathbf{W}^{(3)\top} \vec{\delta}^{z(3)} \\ \vec{\delta}^{z(1)} &= \tilde{g}'(\vec{z}^{(1)}) \odot \vec{\delta}^{a(1)} & \vec{\delta}^{z(2)} &= \tilde{g}'(\vec{z}^{(2)}) \odot \vec{\delta}^{a(2)} & \vec{\delta}^{z(3)} &= \left[\frac{\partial l(y_i, f)}{\partial f} \right] \end{aligned}$$

Backward Propagation

$$\vec{x} = \vec{a}^{(0)} \quad \mathbf{W}^{(1)} \vec{a}^{(0)} = \vec{z}^{(1)} \quad \tilde{g}(\vec{z}^{(1)}) = \vec{a}^{(1)} \quad \mathbf{W}^{(2)} \vec{a}^{(1)} = \vec{z}^{(2)} \quad \tilde{g}(\vec{z}^{(2)}) = \vec{a}^{(2)} \quad \mathbf{W}^{(3)} \vec{a}^{(2)} = \vec{z}^{(3)} \quad \vec{z}^{(3)} = f$$



$$\vec{\delta}^{a^{(0)}} = \mathbf{W}^{(1)\top} \vec{\delta}^{z^{(1)}}$$

$$\vec{\delta}^{a^{(1)}} = \mathbf{W}^{(2)\top} \vec{\delta}^{z^{(2)}}$$

$$\vec{\delta}^{a^{(2)}} = \mathbf{W}^{(3)\top} \vec{\delta}^{z^{(3)}}$$

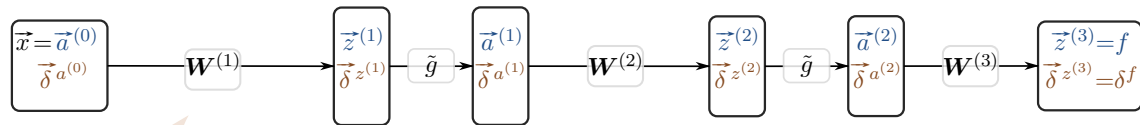
$$\vec{\delta}^{z^{(1)}} = \tilde{g}'(\vec{z}^{(1)}) \odot \vec{\delta}^{a^{(1)}}$$

$$\vec{\delta}^{z^{(2)}} = \tilde{g}'(\vec{z}^{(2)}) \odot \vec{\delta}^{a^{(2)}}$$

$$\vec{\delta}^{z^{(3)}} = \left[\frac{\partial l(y_i, f)}{\partial f} \right]$$

Backward Propagation

$$\vec{x} = \vec{a}^{(0)} \quad \mathbf{W}^{(1)} \vec{a}^{(0)} = \vec{z}^{(1)} \quad \tilde{g}(\vec{z}^{(1)}) = \vec{a}^{(1)} \quad \mathbf{W}^{(2)} \vec{a}^{(1)} = \vec{z}^{(2)} \quad \tilde{g}(\vec{z}^{(2)}) = \vec{a}^{(2)} \quad \mathbf{W}^{(3)} \vec{a}^{(2)} = \vec{z}^{(3)} \quad \vec{z}^{(3)} = f$$



$$\vec{\delta}^{a^{(0)}} = \mathbf{W}^{(1)\top} \vec{\delta}^{z^{(1)}}$$

$$\vec{\delta}^{a^{(1)}} = \mathbf{W}^{(2)\top} \vec{\delta}^{z^{(2)}}$$

$$\vec{\delta}^{a^{(2)}} = \mathbf{W}^{(3)\top} \vec{\delta}^{z^{(3)}}$$

$$\vec{\delta}^{z^{(1)}} = \tilde{g}'(\vec{z}^{(1)}) \odot \vec{\delta}^{a^{(1)}}$$

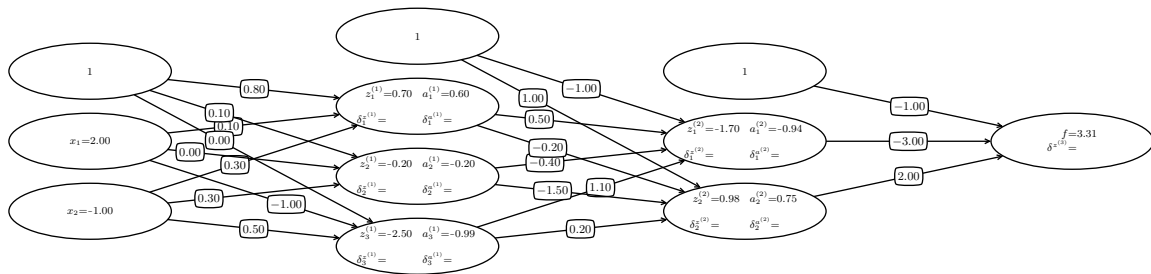
$$\vec{\delta}^{z^{(2)}} = \tilde{g}'(\vec{z}^{(2)}) \odot \vec{\delta}^{a^{(2)}}$$

$$\vec{\delta}^{z^{(3)}} = \left[\frac{\partial l(y_i, f)}{\partial f} \right]$$

backward propagation (calculates $\nabla_{\vec{z}^{(k)}}(l)$)

Backward Propagation Example

$$\begin{aligned}
 \mathbf{W}^{(1)} &= \begin{bmatrix} 0.80 & 0.10 & 0.30 \\ 0.10 & 0.00 & 0.30 \\ 0.00 & -1.00 & 0.50 \end{bmatrix} & \mathbf{z}^{(1)} = \mathbf{a}^{(1)} = \begin{bmatrix} 0.70 \\ -0.20 \\ -2.50 \end{bmatrix} & \mathbf{W}^{(2)} = \begin{bmatrix} -1.00 & 0.50 & -0.40 & 1.10 \\ 1.00 & -0.20 & -1.50 & 0.20 \end{bmatrix} & \mathbf{z}^{(2)} = \mathbf{a}^{(2)} = \begin{bmatrix} -1.70 \\ 0.98 \end{bmatrix} & \mathbf{W}^{(3)} = [-1.00 \ -3.00 \ 2.00] & y=3.0 \\
 \delta^{z^{(1)}} = \delta^{a^{(1)}} & & & \delta^{z^{(2)}} = \delta^{a^{(2)}} & & \delta^{z^{(3)}} =
 \end{aligned}$$

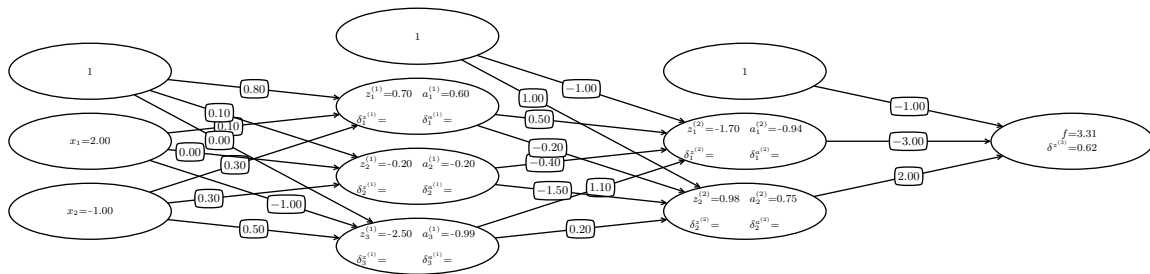


$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

Backward Propagation Example

$$\begin{aligned}
 \mathbf{W}^{(1)} &= \begin{bmatrix} 0.80 & 0.10 & 0.30 \\ 0.10 & 0.00 & 0.30 \\ 0.00 & -1.00 & 0.50 \end{bmatrix} & \mathbf{z}^{(1)} = \mathbf{a}^{(1)} = \begin{bmatrix} 0.70 \\ -0.20 \\ -2.50 \end{bmatrix} & \mathbf{W}^{(2)} = \begin{bmatrix} -1.00 & 0.50 & -0.40 & 1.10 \\ 1.00 & -0.20 & -1.50 & 0.20 \end{bmatrix} & \mathbf{z}^{(2)} = \mathbf{a}^{(2)} = \begin{bmatrix} 1.00 \\ -1.70 \\ 0.98 \end{bmatrix} & \mathbf{W}^{(3)} = [-1.00 \quad -3.00 \quad 2.00] & y = 3.0 \\
 \delta^{z^{(1)}} = \delta^{a^{(1)}} & & & \delta^{z^{(2)}} = \delta^{a^{(2)}} & & \delta^{z^{(3)}} = \begin{bmatrix} 0.62 \end{bmatrix}
 \end{aligned}$$

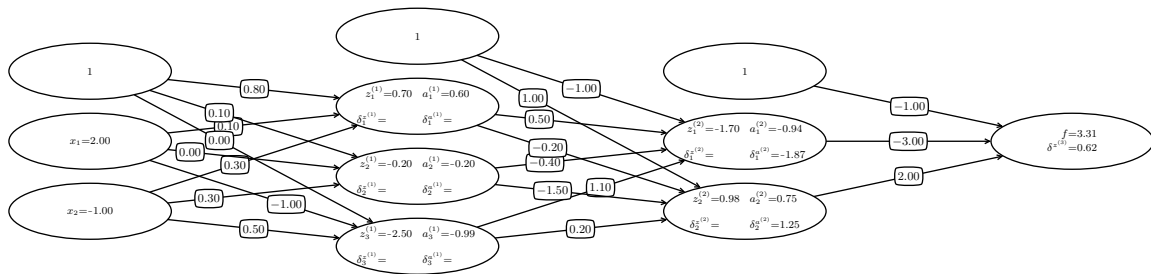


$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

Backward Propagation Example

$$\begin{aligned}
 \mathbf{W}^{(1)} &= \begin{bmatrix} 0.80 & 0.10 & 0.30 \\ 0.10 & 0.00 & 0.30 \\ 0.00 & -1.00 & 0.50 \end{bmatrix} & \mathbf{z}^{(1)} = \mathbf{a}^{(1)} & \mathbf{W}^{(2)} &= \begin{bmatrix} -1.00 & 0.50 & -0.40 & 1.10 \\ 1.00 & -0.20 & -1.50 & 0.20 \end{bmatrix} & \mathbf{z}^{(2)} = \mathbf{a}^{(2)} & \mathbf{W}^{(3)} &= [-1.00 \quad -3.00 \quad 2.00] \\
 \delta^{z^{(1)}} &= \delta^{a^{(1)}} & \delta^{z^{(2)}} &= \delta^{a^{(2)}} & \delta^{z^{(3)}} &= \delta^{a^{(3)}}
 \end{aligned}$$

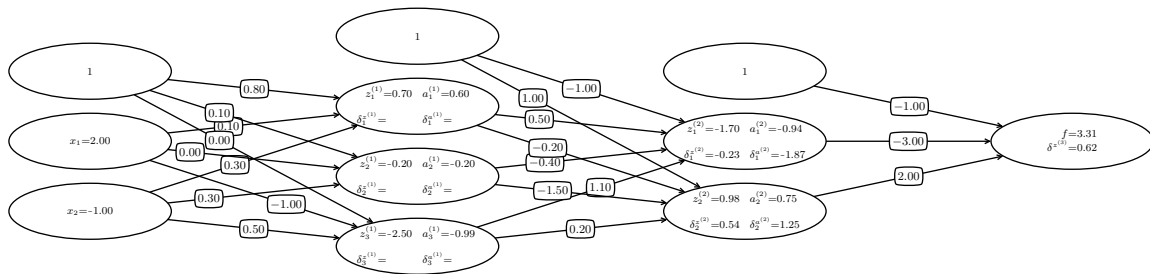


$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

Backward Propagation Example

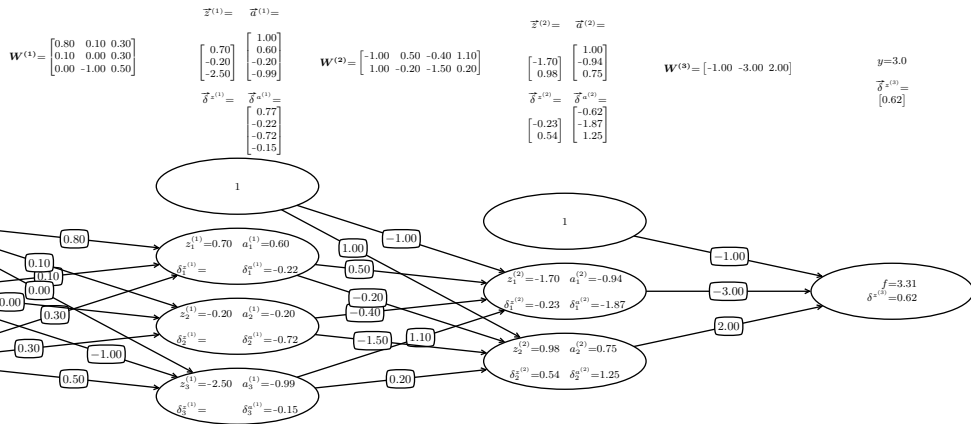
$$\begin{aligned}
 \mathbf{W}^{(1)} &= \begin{bmatrix} 0.80 & 0.10 & 0.30 \\ 0.10 & 0.00 & 0.30 \\ 0.00 & -1.00 & 0.50 \end{bmatrix} & \mathbf{z}^{(1)} = \begin{bmatrix} 0.70 \\ -0.20 \\ -2.50 \end{bmatrix} & \mathbf{a}^{(1)} = \begin{bmatrix} 1.00 \\ 0.60 \\ -0.20 \\ -0.99 \end{bmatrix} & \mathbf{W}^{(2)} = \begin{bmatrix} -1.00 & 0.50 & -0.40 & 1.10 \\ 1.00 & -0.20 & -1.50 & 0.20 \end{bmatrix} & \mathbf{z}^{(2)} = \begin{bmatrix} -1.70 \\ 0.98 \end{bmatrix} & \mathbf{a}^{(2)} = \begin{bmatrix} 1.00 \\ -0.94 \\ 0.75 \end{bmatrix} & \mathbf{W}^{(3)} = [-1.00 \ -3.00 \ 2.00] & y = 3.0 \\
 \delta^{z^{(1)}} = \delta^{a^{(1)}} & & \delta^{z^{(2)}} = \delta^{a^{(2)}} & & \delta^{z^{(3)}} = \delta^{a^{(3)}} & & & & \delta^{z^{(3)}} = [0.62]
 \end{aligned}$$



$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

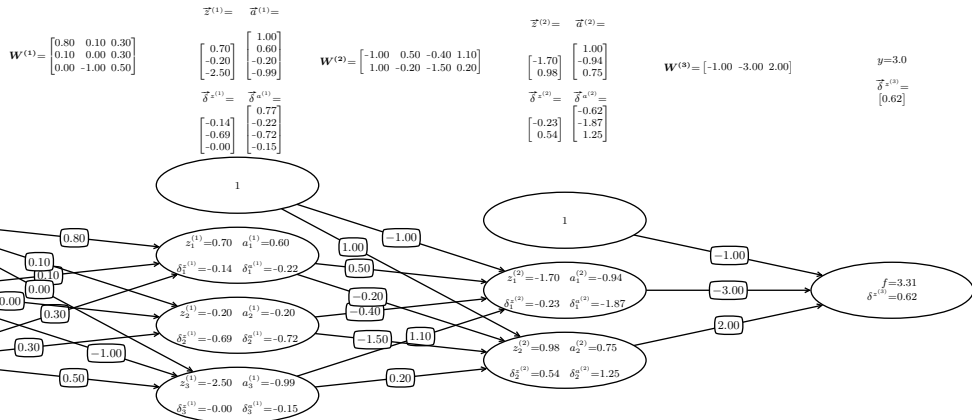
Backward Propagation Example



$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

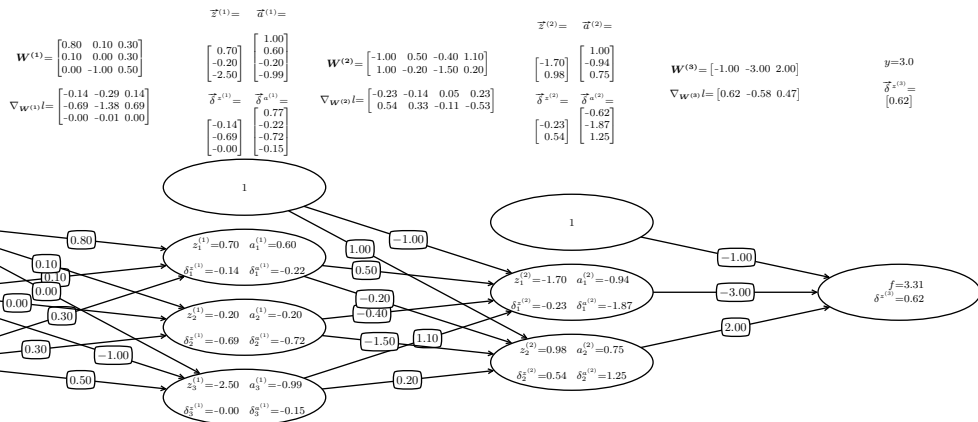
Backward Propagation Example



$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

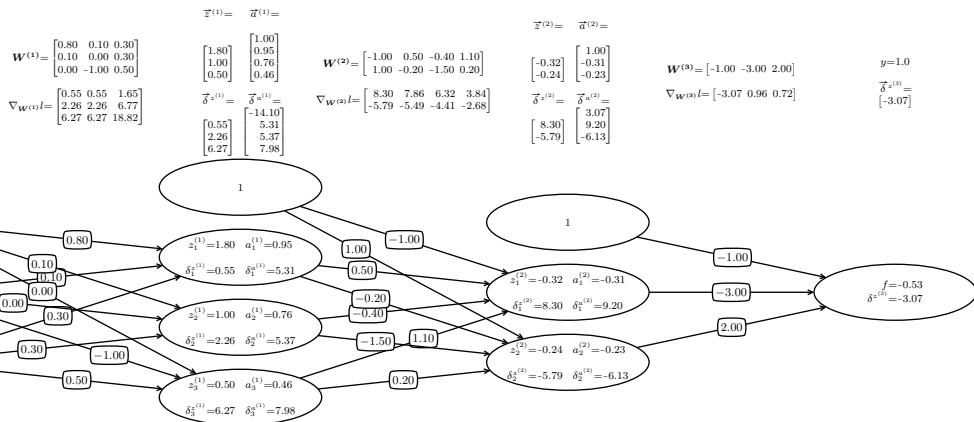
Backward Propagation Example



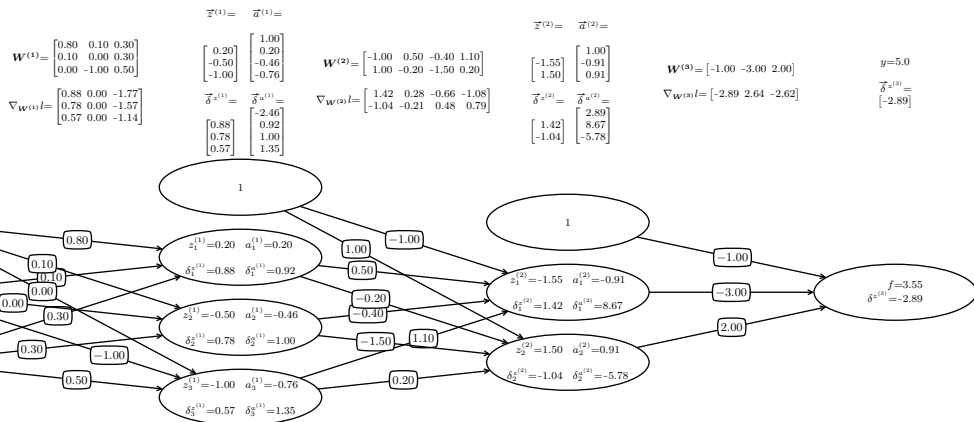
$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$

Backward Propagation Example



Backward Propagation Example



$$g(z) = \tanh(z)$$

$$l(y, f) = (f - y)^2$$